Matthew Dong

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Summary

Top 10% Master's graduate at **New York University** with **3+** years of **Python** and **PyTorch** experience, specializing in **DevOps** and **CI/CD** automation using **GitHub Actions** to streamline code pipelines. Experienced in cloud computing with **Google Cloud Platform (GCP)**. Proficient in fine-tuning open-source transformer **LLMs** to personalize them and training models to avoid hallucination by using **RAG**. Well-versed in containerization and orchestration using **Docker** and **Kubernetes (K8s)**, as well as **PostgreSQL** data visualization tools such as **Domo** and **Tableau**.

Education

New York University — **New York, NY**
Master of Science: Computer Science — **GPA: 3.95/4.00** — Sep 2024 – May 2025

New York University — **New York, NY**
Bachelor of Arts: Computer Science; Bachelor of Arts: Data Science — **GPA: 3.93/4.00** — Sep 2021 – May 2024

Experience

NJStat — **Bridgewater, NJ**
Artificial Intelligence Intern — Jan 2025 – Present

- Updated **natural language processing** AI using the **Transformers** library, improved employee monitoring by 20%.
- Leveraged **zero-shot** and **chain-of-thought prompt engineering** techniques in lieu of fine-tuning to overcome limited computing resources and drive project success.
- Demonstrated the value of developed applications to management with **Figma** prototypes, accelerating project approvals and maintaining consistent communication between the team and management.
- Revamped the company timesheet application by designing a robust **SQL** database schema, processing data, implementing **full-stack** (**front-end** and **back-end**) functionality, and authoring comprehensive **documentation** for future maintenance.

Colgate-Palmolive — **Piscataway, NJ**
Application and Software Development Intern (E-Commerce DevOps) — Jun 2022 – Dec 2024

- Developed **Python** scripts that utilize **cloud functions** and **API gateways** to parse **JSON** alerts for various applications and send them to Google Chat spaces using **webhooks**, clearing cluttered email inboxes.
- Automated code updates by implementing robust **CI/CD** workflows using **GitHub Actions** while also maximizing reusability—deploying **Python** scripts to **Google Cloud Platform** as **cloud functions** that interface with **API gateways.**
- Streamlined **Splunk** panels to convey key information about company servers to developers in an easy to view dashboard.
- Investigated the benefits of using **computer vision** models such as **DALL-E 3** for creating internal company diagrams by implementing a prototype with the **OpenAI API**.

Sanofi — **Bridgewater, NJ**
Data Analyst Intern — Dec 2020 – Feb 2021

- Applied **PCA**, **random forests**, and **K-means clustering** to analyze and extract insights from high-dimensional financial and economic data.
- Utilized **R** to analyze **COVID-19** data, applying linear, logistic, and multiple **regression** techniques to derive insights.

Personal Projects

(More projects available on GitHub: github.com/Matt-J-Dong)

VentureAI (Python: PyTorch)

- Developed an automated travel agent that utilizes fine-tuned **open-source LLMs** that provides personalized travel recommendations tailored to each user's preferences by integrating the LLM with APIs for dynamic **retrieval augmented generation** (Dynamic RAG)
- Addressed limitations of state of the art models (hallucination) and implemented innovations such as **LoRA** and **quantization**.

Multicore Chess (C++: OpenMP)

- Greatly improved the speed of move calculations, achieving up to **12x** speedup in comparison to the sequential version.
- Optimized the evaluation function and **parallel alpha-beta pruning** algorithms despite limited CPUs and no access to GPUs.

Top-Towns-To-Take-Over-Tech (Spark Scala, Java MapReduce, Shell)

- Built an analytic with a group to find out which American cities are the best for tech careers.
- We cleaned, profiled, and performed ETL on our data before joining all the datasets together, and displayed our results through a Tableau visualization and a Hive warehouse.

Skills

Languages: Python, R, Markdown, C++, Spark Scala, MySQL, R, Bash, Java, x86-64 Assembly
Libraries: PyTorch, Transformers, Wandb, NumPy, Pandas, Matplotlib, Seaborn, Scikit-Learn, OpenMP, MPI
Technologies: GitHub, GitHub Actions, Hadoop, Kubernetes (K8s), Docker, GCP, Terraform, Jira, Confluence, Regex, HTML, Splunk, Linux, MongoDB, Figma